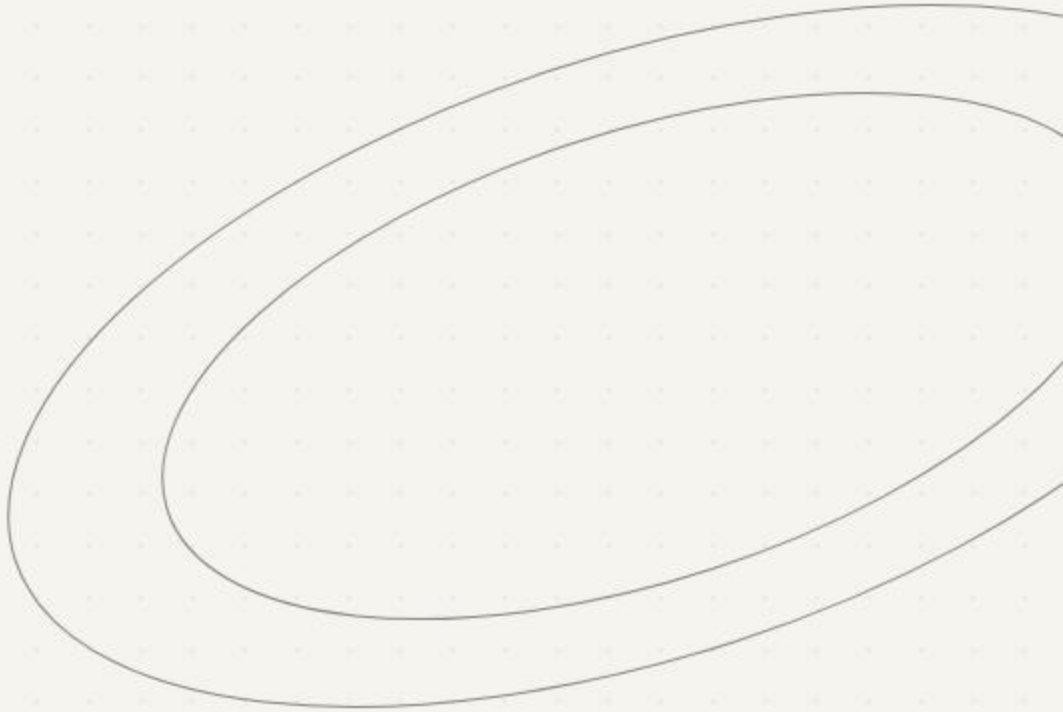
CELESTIAL MECHANICS × AI · 2026

From the next token to the next theorem

How a model works — and how I use it in practice



Bowen Liu

Overview: from token to theorem

Click a section card to jump; click a tag for a key page.

00 · OPENING + TOKEN

Token

Ten integers · vocabulary · billing

p. 3–8

Ten integers · Vocabulary

01 · MODEL

One function

Distribution

p. 9

Distribution

02 · TRAINING & PROBLEM SOLVING

Learning from reward

RL loop · math and code · GRPO

p. 10–23

RL loop · GRPO formula · REFLEX · Stateless

03 · SYSTEMS & ENGINEERING

Put the model in a system

Context · harness · skills

p. 24–30

Harness · System map · Skill

04 · AI × MATH

From tools to search

Acceleration · timeline · pattern

p. 31–36

Timeline · Pattern

05 · CONCLUSION

Back to practice

Workflow · compounding · close

p. 37–38

Close

6 sections · 38 pages · Click any card to start; `N` notes, `F` fullscreen.

REFLEX: search engine in the RL loop

E-commerce query expansion: generate → retrieve → reward result quality.

Query + top-k products → LLM expands → Search engine → Union of results → Reward ↺

REWARD · NO HUMAN LABELS

Relevance Rel(Y)/Rel(q) + coverage Ret(Y)/Ret(q); format violations score 0.

TRAINING

SFT warm start on 10k → GRPO on 100k real queries; 8×H200.

QUERIES WITH POSITIVE GAIN · %

SETTING	ΔCOV	ΔREL
SFT ONLY	12.8	8.9
GRPO ONLY	57.5	25.2
WARM START + GRPO	72.8	32.4

10,000 queries · 5 samples · LLaMA 3.2-3B

> GRPO turns **better retrieval** into an optimizable reward.

Zhang, Liu, et al., REFLEX, WSDM 2026 · ablation on this dataset and backbone.

After training: four limits, three system capabilities

The model predicts; state, computation and reliability live outside it.

01 · MEMORY & CONTEXT

No memory, limited window

- Parameters fixed at inference; knowledge in weights
- A "conversation" resends history
- 4k → 1M, but long inputs still lose the middle

→ Context management · retrieval · memory

02 · EXACT COMPUTATION

Generated, not executed

- Arithmetic is token prediction, not calculation
- Plausible answers can still be wrong
- Errors compound over long tasks

→ Function calling · call a program

03 · PROBABILISTIC OUTPUT

Same input, different output

- Output is sampled, not a deterministic function
- Errors can sound confident and plausible
- Long runs need stopping and rollback

→ Harness · verification · permissions · evals

> Four limits, three system capabilities: **state, computation, reliability**.

Context and memory: a finite window, used well

The window is a budget, not a warehouse: place the right things, then decide what stays.

01 · FINITE WINDOW

4k → 1M, still lossy

Needle in a haystack: the middle is diluted first.



Illustrative · Liu et al., 2023

Capacity ≠ effective use.

02 · PLACEMENT

Workbench, not warehouse

system · rules

history · recent turns

retrieval · relevant docs

tools · returned results

current · this question

Too long→retrieve; too scattered→structure; overflow→summarize or sub-agents.

03 · RETENTION

Signal, not logs

Signal

tasks · constraints · decisions · facts

Noise

chat · repetition · stale · raw logs

WRITE
decisions and facts only

RETRIEVE
for this task

COMPRESS
summarize, merge

FORGET
delete stale

working · episodic · semantic · procedural

> Capacity ≠ effective use; Context decides **what goes in**, Memory decides **what stays**.

Tool calling: let a program do the arithmetic

tool schema

```
{
  "name": "run_python",
  "args": {"code": "str"},
  "returns": "str",
  "limits": ["sandbox", "timeout"]
}
```

one call

```
call  run_python("print(2 * 4**1.5)")
obs   "16.0"
```

```
// T = 2 × 4**1.5 = 16
```

calc · code · search · database · files · APIs

01 DEFINE

Name, arguments, return, boundary; the description is context too.

02 CALL

The model emits a tool_call; it proposes, does not execute.

03 EXECUTE

The runtime validates arguments and runs in a sandbox; timeouts and permissions are system concerns.

04 OBSERVE

stdout / stderr return to the context; the model decides to continue or stop.

> Tool output is still **untrusted input**: validate arguments, limit permissions, set timeouts, keep logs.

> The model proposes, the program executes, **the system owns the boundary**.

Harness: turn probabilistic output into a runnable system

The model proposes; the harness makes the proposal legal, executable, checkable, and stoppable.

A controlled loop

Propose → Validate → Execute → Observe → Verify → Decide

WHAT THE HARNESS OWNS

Tool registry and argument validation · sandbox and permissions · timeouts and retries · result verification · rollback and stop · logs and replay

> It does not add intelligence; it makes errors **visible, stoppable, reproducible.**

01 PERMISSIONS & SANDBOX

What may run, what may not; constrain before execution

02 TIMEOUTS & RETRIES

Failure must not loop forever; retries have a limit

03 VERIFICATION & ROLLBACK

Is the result right? Can a bad change be undone?

04 LOGS & STOPPING

Can it be reproduced? When must it stop?

> The model provides probability; the harness provides the **boundary**.

Put the model back in the system



Inner ring: one call · Middle ring: across calls · Outer ring: engineering boundary

01	**LLM**	A probability generator: tokens in, a distribution over the next token out
02	**Prompt**	This call's conditions: system / user / few-shot
03	**Context**	Every token visible to this call; the window is finite
04	**Memory**	State beyond the window: write / retrieve / compress / forget
05	**Agent loop**	Observe → act → observe; state lives outside
06	**Harness**	Tools, sandbox, permissions, verification, logs

> Memory owns state, tool calls compute, harness verifies: together they form an **agent system**.

Skill: package a procedure into a reusable capability

Not a bigger model; a load-on-demand bundle of steps, scripts, and references.

```
skill package
skill/
  SKILL.md        when · steps · checks
  scripts/        deterministic ops
  references/     domain docs
  assets/         templates
```

Outline → Generate → Screenshot → Audit → Export

Example: this deck's production loop as a skill

01 TRIGGER

Name and description: when to use it, when not to

02 STEPS

SKILL.md: what comes first, what follows, where the checks are

03 TOOLS

scripts/: freeze deterministic operations; stop rewriting them

04 ASSETS

references/ and assets/: domain knowledge, templates, examples

A skill is loaded on demand, versioned, and reusable; context budget and tool permissions still apply.

Turn a one-off prompt into a **versioned procedure**.

From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

People set the direction

> The next theorem still begins with the **next token**.